Ishan Sharma

Software Engineer (Backend / Full-Stack) | Final-Year B.E. Computer Science Engineering
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github.com/AlfredBateman | alfredbateman.github.io/portfolio-v3

SUMMARY

Final-year Computer Science Engineering student seeking a Software Engineer (Backend / Full-Stack) role. Built and deployed three full-stack projects using Python, FastAPI, React, Next.js, Node.js and PostgreSQL, with REST APIs, CI/CD pipelines (GitHub Actions), Docker and 1,000+ automated tests. Strong in data structures and algorithms (DSA), system design and testing.

TECHNICAL SKILLS

- **Languages:** Python, TypeScript, JavaScript, Java, C++, C, SQL
- **Backend & Databases:** FastAPI, Node.js, Express.js, REST APIs, PostgreSQL, MongoDB, SQLite, Prisma ORM
- **Frontend:** React, Next.js, Zustand, Vite, Tailwind CSS, Chrome Extensions (Manifest V3)
- **DevOps & Testing:** CI/CD, GitHub Actions, Docker, Linux, AWS, Git, GitHub, unit testing, integration testing, end-to-end testing (Playwright), Vitest, pytest
- **AI / LLM:** Gemini API, LiteLLM, Model Context Protocol (MCP), LLM-agent evaluation, schema validation (Zod)
- **Core Concepts:** Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), System Design, Database Management Systems (DBMS), Operating Systems

PROJECTS

AgentProbe: Regression Testing for LLM Agents **github.com/AlfredBateman/AgentProbe**
Python, FastAPI, Next.js, PostgreSQL, Docker, GitHub Actions agent-probe-umber.vercel.app

- Built a regression-testing platform for LLM agents (Python, FastAPI, REST API): each case runs repeatedly (10 times in the shipped suites) and is judged on the full trace and tool calls by rule-based, LLM-rubric and consistency judges.
- Implemented the verdict statistics (per-case Fisher exact test with Tarone–Holm correction plus a paired sign-flip permutation test) in exact arithmetic; in simulation, false regression alarms on unchanged flaky agents fell from 70.6% to 0.9%.
- Shipped a FastAPI and PostgreSQL backend with JWT authentication, a Next.js dashboard, a CLI, and a GitHub Actions integration that comments on pull requests and fails the build on a regression; added an SSRF guard.
- Backed by 1,145 pytest tests, 184 web unit tests and 14 Playwright end-to-end specs (97% coverage on core and API), with a CI/CD pipeline deploying to Vercel, Render and Neon.

StockPilot: Stock Screener with Natural-Language Queries **github.com/AlfredBateman/StockPilot**
React, Node.js, Express, TypeScript, MongoDB Atlas, Render, GitHub Actions stockpilot-hw7d.onrender.com

- As team lead of 3, built and deployed a full-stack screener over about 100 NIFTY stocks (filter, sort, search, watchlist, 3-way compare, price-history metrics); a weekday GitHub Actions job refreshes end-of-day Yahoo Finance data into MongoDB Atlas, with a committed snapshot as fallback.
- Designed a plain-English query pipeline (typo fixing, offline rule parser, company-name search, hosted LLM, answer cache) with the deterministic parser as the floor, so search never depends on a network call; model output is schema-validated.
- Added "why this matched" explanations on every result row, a beginner glossary, a Guide drawer, and zero-result suggestions that show which filter to drop; released v1.2.0.
- Wrote 470+ Vitest unit tests and an 81-query evaluation harness (rule parser matches 80 of 81 self-authored expectations).

CanvasLint: Canvas 2D Debugger for Chrome DevTools **github.com/AlfredBateman/CanvasLint**
TypeScript, React, Zustand, Vite, Chrome Manifest V3 v1.1.0

- Built a Manifest V3 extension that records Canvas 2D calls, batches them per frame, and keeps a capped per-tab history (10,000 frames or about 32 MiB) in a service worker that validates every message from the page.
- Built a DevTools panel with a scrubbable timeline that replays the command log (chosen over pixel snapshots for exact replay), a state inspector, a seven-rule lint engine, and export/import of recordings.
- Cut worker CPU per page frame from 11.8 ms to under 2 ms, with about 9% frame-time overhead on a 7,034-command benchmark page; added keyboard control and an end-to-end privacy test; released v1.1.0.

EDUCATION

Chitkara University **Expected Aug 2027**
Bachelor of Engineering, Computer Science Engineering Himachal Pradesh, India

- Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, System Design, Computer Networks

CERTIFICATIONS

- AWS Cloud Practitioner Essentials; AWS Cloud Quest: Cloud Practitioner; Nutanix Hybrid Cloud Fundamentals (NHCF); Infosys Springboard: Python Foundation, Software Development & Testing Fundamentals, Programming Fundamentals using Python